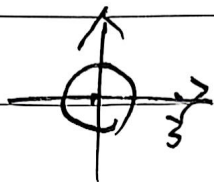


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8.12)

$$\oint_{|z|=1} \frac{z^2 + e^z}{z^3} dz = 2\pi i \left. \frac{z^2 + e^z}{z^3} \right|_{z=1} = -\pi i(1+e)$$



$$13) \oint \frac{z^2-1}{(z+i)(z-i)} dz = \frac{1}{2i} \left(\frac{1}{z-i} - \frac{1}{z+i} \right) (z^2+1) dz$$

$$= \frac{1}{2i} \left(\oint \frac{z^2+1}{z-i} dz + \oint \frac{z^2+1}{z+i} dz \right)$$

$$= \frac{1}{2i} \cdot (2\pi i (z^2+1)|_{z=i} + 2\pi i (z^2+1)|_{z=-i}) = \cancel{1} \quad 0$$

9. ① 若 z_0 在 $\mathbb{C}$ 外:

则 $LHS = RHS = 0$ 成立

② 若 z_0 在 $\mathbb{C}$ 内

则 $LHS = 2\pi i f'(z)|_{z=z_0}$

$RHS = \frac{2\pi i}{1!} f'(z)|_{z=0}$

$\therefore LHS = RHS$ 成立

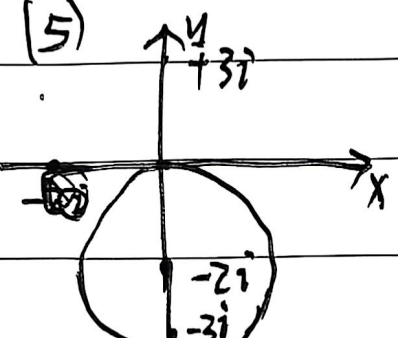
$\therefore$ 总的, 恒成立

12 (2)

$$LHS = 2\pi i (\sin z)' \Big|_{z=\frac{\pi}{2}} = 0$$

(4)  $LHS = \frac{2\pi i}{2!} (e^{tz})'' \Big|_{z=0} = \pi i \cdot t^2$

(3) $LHS = \frac{2\pi i}{3!} (\sin \pi z)^{(3)} \Big|_{z=1} = \frac{\pi i}{3} \pi^3 (-\cos \pi z) \Big|_{z=1} = -i \frac{\pi^4}{3}$

(5) 
$$LHS = \oint \frac{1}{(z-3i)^2} dz = \frac{2\pi i}{1!} \left(\frac{1}{(z-3i)^2} \right)' \Big|_{z=-3i}$$

$$= 2\pi i \cdot 2 \cdot \frac{1}{(z-3i)^3} \Big|_{z=-3i} = -\frac{\pi}{54} i$$

(6) $LHS = \frac{2\pi i}{2!} (5z^2 - 3z + 2)'' \Big|_{z=1} = 10\pi i$

14. $f(z)$ 在 $\mathbb{C}$ 上解析,

C-M: $\oint \frac{f(z)}{(z-z_0)^2} dz = 2\pi i f'(z_0)$

$$f'(z_0) = \frac{1}{2\pi i} \oint \frac{f(z)}{(z-z_0)^2} dz = \frac{1}{2\pi i} \oint \frac{\operatorname{Re} f(z) + i \operatorname{Im} f(z)}{(z-z_0)^2} dz$$

$$< \frac{1}{2\pi i} \oint \frac{M + i \operatorname{Im} f(z)}{(z-z_0)^2} dz = \frac{1}{2\pi i} \left[M \frac{2\pi R}{R^2} + i \oint \frac{\operatorname{Im} f(z)}{R^2} dz \right]$$

设 $F(z) = e^{f(z)} = e^{\operatorname{Re} f(z)} < e^M$

$\Rightarrow F(z)$ 为常数 $\Rightarrow f(z)$ 为常数

构造有界函数